

Supplementary File S6

RQ6: Reproducibility, Benchmarking, and Research Infrastructure

Supporting material for *What the Survey Literature Says About Reinforcement Learning for Building Control: A Critical Synthesis*

Scope and coding scale

This supplementary file provides the survey-level evidence audit for RQ6. It reports infrastructure coverage, reproducibility and comparability critiques, recurring infrastructure gaps, and concise survey-level answers.

Score	Evidence-intensity interpretation
0	Not addressed
1	Mentioned
2	Discussed
3	Structuring role

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S6.1 A review of reinforcement learning for autonomous building energy management [1]

Table S6.1: RQ6 recoded audit table for Mason and Grijalva (2019). Scores use the shared 0–3 evidence-intensity scale.

Survey	Reproducibility auditing	Infrastructure auditing	Blind-spot visibility	Recoded evidence justification
Mason and Grijalva (2019)	1	2	1	The review is infrastructure-aware but only lightly reproducibility-critical. Reproducibility auditing receives score 1 because the paper does not systematically audit open code, benchmark comparability, dataset openness, artifact availability, or reporting standards. Infrastructure auditing receives score 2 because the paper discusses the wider technical stack needed for autonomous building energy management, including sensors, actuators, communication infrastructure, IoT, smart meters, batteries, smart-grid integration, simulator design, electricity prices, weather, PV variability, and occupancy-related state information. Blind-spot visibility receives score 1 because simulator design and transfer learning are identified as concerns for scaling RL beyond single-building controller training, but the paper does not develop these issues as a strong reproducibility or shared-infrastructure agenda.

Table S6.2: RQ6 infrastructure-coverage scores for Mason and Grijalva (2019), recoded using the shared evidence-intensity scale.

Survey	Simulators / environments	Datasets / data sources	Benchmark tasks	Code / artifact availability	External signals	Reporting / experimental detail
Mason and Grijalva (2019)	2	0	0	0	2	1

Table S6.3: RQ6 critical-audit scores for Mason and Grijalva (2019), recoded using the shared evidence-intensity scale.

Survey	Benchmark comparability critique	Open-code / reproducibility critique	Simulator fragmentation critique	Data realism / availability critique	Standardization / shared-infrastructure need
Mason and Grijalva (2019)	0	0	1	0	0

Table S6.4: RQ6 recurring infrastructure-gap scores for Mason and Grijalva (2019), recoded using the shared evidence-intensity scale.

Survey	Closed code / unavailable artifacts	Lack of shared benchmarks	Inconsistent external signals / assumptions	Simulator / testbed fragmentation	Weak dataset realism / documentation
Mason and Grijalva (2019)	0	0	0	1	0

Table S6.5: RQ6 summary table for Mason and Grijalva (2019), after recoding.

Survey	Reproducibility orientation	Dominant infrastructure concern	Main RQ6 consequence
Mason and Grijalva (2019)	Infrastructure-aware but only lightly reproducibility-critical	The survey’s main infrastructure concern is that reinforcement-learning-based building control depends on a wider technical stack, especially simulator design, smart-metering, IoT connectivity, and grid-linked external information such as prices and weather.	The survey contributes to RQ6 mainly by showing that reinforcement learning for building energy management rests on nontrivial technical infrastructure and external-signal assumptions, but it does not yet develop a strong audit of reproducibility, benchmark comparability, open artifacts, or dataset standardization.

Table S6.6: One-line RQ6 answer for Mason and Grijalva (2019), after recoding.

Survey	RQ6 answer
Mason and Grijalva (2019)	Mason and Grijalva (2019) address RQ6 mainly by showing that reinforcement-learning-based building energy management depends on simulators, IoT-connected devices, smart meters, and external signals such as weather and pricing, while only lightly auditing reproducibility itself and giving little direct attention to open code, shared benchmarks, or dataset standardization.

S6.2 Reinforcement learning for building controls: The opportunities and challenges [2]

Table S6.7: RQ6 recoded audit table for Wang and Hong (2020). Scores use the shared 0–3 evidence-intensity scale.

Survey	Reproducibility auditing	Infrastructure auditing	Blind-spot visibility	Recoded evidence justification
Wang and Hong (2020)	3	3	3	The review gives a strong infrastructure and reproducibility critique. Reproducibility auditing receives score 3 because the paper structurally audits training-data sources, environmental settings, simulator dependence, real-building implementation, comparison baselines, and cross-study comparability. Infrastructure auditing receives score 3 because the paper discusses simulators, actual buildings, open-source control testbeds, Open Building Controls, CityLearn, external signals, and shared datasets as core infrastructure for RL-based building control. Blind-spot visibility receives score 3 because the paper explicitly argues that fair comparison is difficult without common, credible benchmarks and calls for open-source and well-recognized control testbeds and datasets.

Table S6.8: RQ6 infrastructure-coverage scores for Wang and Hong (2020), recoded using the shared evidence-intensity scale.

Survey	Simulators / environments	Datasets / data sources	Benchmark tasks	Code / artifact availability	External signals	Reporting / experimental detail
Wang and Hong (2020)	3	1	3	1	2	2

Table S6.9: RQ6 critical-audit scores for Wang and Hong (2020), recoded using the shared evidence-intensity scale.

Survey	Benchmark comparability critique	Open-code / reproducibility critique	Simulator fragmentation critique	Data realism / availability critique	Standardization / shared-infrastructure need
Wang and Hong (2020)	3	1	2	2	3

Table S6.10: RQ6 recurring infrastructure-gap scores for Wang and Hong (2020), recoded using the shared evidence-intensity scale.

Survey	Closed code / unavailable artifacts	Lack of shared benchmarks	Inconsistent external signals / assumptions	Simulator / testbed fragmentation	Weak dataset realism / documentation
Wang and Hong (2020)	0	3	2	2	1

Table S6.11: RQ6 summary table for Wang and Hong (2020), after recoding.

Survey	Reproducibility orientation	Dominant infrastructure concern	Main RQ6 consequence
Wang and Hong (2020)	Benchmark- and infrastructure-aware, with a clear reproducibility critique	The survey’s dominant infrastructure concern is the lack of common and credible benchmarking infrastructure, given the heavy dependence on simulators, inconsistent comparison baselines, weak real-building adoption, and the need for open-source testbeds and datasets.	The survey contributes strongly to RQ6 by showing that reinforcement-learning-based building-control research is not only simulation-heavy, but also difficult to compare fairly across studies without shared evaluation environments, better external-signal handling, and more standardized benchmarking practices.

Table S6.12: One-line RQ6 answer for Wang and Hong (2020), after recoding.

Survey	RQ6 answer
Wang and Hong (2020)	Wang and Hong (2020) answer RQ6 by explicitly showing that reinforcement-learning-based building-control research depends heavily on simulators, lacks shared and credible benchmarks for fair comparison, underuses important external signals such as forecasts, and therefore needs open-source datasets and standardized testbeds to improve reproducibility and infrastructure quality.

S6.3 The reinforcement learning method for occupant behavior in building control: A review [3]

Table S6.13: RQ6 recoded audit table for Han et al. (2021). Scores use the shared 0–3 evidence-intensity scale.

Survey	Reproducibility auditing	Infrastructure auditing	Blind-spot visibility	Recoded evidence justification
Han et al. (2021)	1	2	2	The review is data- and infrastructure-aware, but only lightly reproducibility-critical. Reproducibility auditing receives score 1 because the paper does not systematically audit open code, artifact availability, shared benchmarks, or reproducible experimental packages. Infrastructure auditing receives score 2 because the paper discusses high-quality data, historical occupant behavior information, repeated occupant feedback, frequent state and reward signals, transfer learning, function approximation, and collaboration for sharing data and research infrastructures. Blind-spot visibility receives score 2 because the paper explicitly identifies weak validation standards, data limitations, context dependence, and the need for stronger collaboration as constraints for occupant-behavior-oriented RL.

Table S6.14: RQ6 infrastructure-coverage scores for Han et al. (2021), recoded using the shared evidence-intensity scale.

Survey	Simulators / environments	Datasets / data sources	Benchmark tasks	Code / artifact availability	External signals	Reporting / experimental detail
Han et al. (2021)	1	3	0	0	1	2

Table S6.15: RQ6 critical-audit scores for Han et al. (2021), recoded using the shared evidence-intensity scale.

Survey	Benchmark comparability critique	Open-code / reproducibility critique	Simulator fragmentation critique	Data realism / availability critique	Standardization / shared-infrastructure need
Han et al. (2021)	0	0	0	2	1

Table S6.16: RQ6 recurring infrastructure-gap scores for Han et al. (2021), recoded using the shared evidence-intensity scale.

Survey	Closed code / unavailable artifacts	Lack of shared benchmarks	Inconsistent external signals / assumptions	Simulator / testbed fragmentation	Weak dataset realism / documentation
Han et al. (2021)	0	0	1	0	2

Table S6.17: RQ6 summary table for Han et al. (2021), after recoding.

Survey	Reproducibility orientation	Dominant infrastructure concern	Main RQ6 consequence
Han et al. (2021)	Data- and infrastructure-aware, but only lightly reproducibility-critical	The survey’s dominant infrastructure concern is the weak standardization of validation practice and the heavy dependence of occupant-centered reinforcement learning on sufficient high-quality data, repeated feedback signals, and collaboration around shared research infrastructure.	The survey contributes to RQ6 mainly by showing that occupant-behavior reinforcement learning still depends on fragile data and validation infrastructures, but it contributes less strongly to benchmark, code, and artifact reproducibility than the more infrastructure-auditing surveys.

Table S6.18: One-line RQ6 answer for Han et al. (2021), after recoding.

Survey	RQ6 answer
Han et al. (2021)	Han et al. (2021) answer RQ6 mainly by showing that reinforcement learning for occupant behavior relies on weakly standardized validation practice, high-quality data, repeated feedback signals, and stronger collaboration around shared knowledge and infrastructures, while giving only limited attention to shared benchmarks, open artifacts, or code reproducibility.

S6.4 A review of deep reinforcement learning for smart building energy management [4]

Table S6.19: RQ6 recoded audit table for Yu et al. (2021). Scores use the shared 0–3 evidence-intensity scale.

Survey	Reproducibility auditing	Infrastructure auditing	Blind-spot visibility	Recoded evidence justification
Yu et al. (2021)	1	2	1	The review is training- and modeling-infrastructure aware, but only lightly reproducibility-critical. Reproducibility auditing receives score 1 because the paper does not strongly audit open code, artifact availability, shared benchmark tasks, or reusable experimental packages. Infrastructure auditing receives score 2 because the paper discusses the dependence of DRL-based smart building energy management on simulations, calibrated environment models, historical operational data, IoT-collected data, cloud computing, big-data analytics, and data-efficient training pipelines. Blind-spot visibility receives score 1 because costly data collection and inefficient direct interaction with real buildings are recognized as barriers, but the paper does not turn these into a strong shared-benchmark or open-artifact agenda.

Table S6.20: RQ6 infrastructure-coverage scores for Yu et al. (2021), recoded using the shared evidence-intensity scale.

Survey	Simulators / environments	Datasets / data sources	Benchmark tasks	Code / artifact availability	External signals	Reporting / experimental detail
Yu et al. (2021)	3	2	1	0	2	3

Table S6.21: RQ6 critical-audit scores for Yu et al. (2021), recoded using the shared evidence-intensity scale.

Survey	Benchmark comparability critique	Open-code / reproducibility critique	Simulator fragmentation critique	Data realism / availability critique	Standardization / shared-infrastructure need
Yu et al. (2021)	1	0	1	2	1

Table S6.22: RQ6 recurring infrastructure-gap scores for Yu et al. (2021), recoded using the shared evidence-intensity scale.

Survey	Closed code / unavailable artifacts	Lack of shared benchmarks	Inconsistent external signals / assumptions	Simulator / testbed fragmentation	Weak dataset realism / documentation
Yu et al. (2021)	0	1	1	1	1

Table S6.23: RQ6 summary table for Yu et al. (2021), after recoding.

Survey	Reproducibility orientation	Dominant infrastructure concern	Main RQ6 consequence
Yu et al. (2021)	Training- and modeling-infrastructure aware, but only moderately reproducibility-critical	The survey’s dominant infrastructure concern is the dependence of DRL-based smart building energy management on environment models, calibrated simulation frameworks, historical or IoT-collected data, and data-efficient training pipelines.	The survey contributes to RQ6 mainly by making the technical infrastructure of DRL-based smart building energy management explicit and by showing that practical applicability depends heavily on data efficiency and environment-model support, but it contributes less strongly to open-code, artifact, and shared-benchmark reproducibility critique.

Table S6.24: One-line RQ6 answer for Yu et al. (2021), after recoding.

Survey	RQ6 answer
Yu et al. (2021)	Yu et al. (2021) answer RQ6 mainly by showing that DRL for smart building energy management relies on substantial modeling and data infrastructure, including simulations, calibrated environment models, and IoT-collected data, while practical scalability is limited by data inefficiency and costly interaction with real buildings rather than by an explicitly developed shared-benchmark or open-artifact culture.

S6.5 A review of reinforcement learning applications to control of heating, ventilation and air conditioning systems [5]

Table S6.25: RQ6 recoded audit table for Sierla et al. (2022). Scores use the shared 0–3 evidence-intensity scale.

Survey	Reproducibility auditing	Infrastructure auditing	Blind-spot visibility	Recoded evidence justification
Sierla et al. (2022)	3	3	3	The review is strongly comparability- and reporting-oriented. Reproducibility auditing receives score 3 because the paper structurally evaluates how action formulation, abstraction level, control targets, and reporting sufficiency affect the comparability and reproducibility of RL-for-HVAC studies. Infrastructure auditing receives score 3 because the review distinguishes physical HVAC systems from virtual environments, including physics-based building simulators and data-driven models, and assesses how these choices affect deployability and interpretation. Blind-spot visibility receives score 3 because unclear actuator or setpoint mappings, excessive abstraction, vague simulator capabilities, and weak links between indoor-environment variables and HVAC action spaces are treated as structural blind spots.

Table S6.26: RQ6 infrastructure-coverage scores for Sierla et al. (2022), recoded using the shared evidence-intensity scale.

Survey	Simulators / environments	Datasets / data sources	Benchmark tasks	Code / artifact availability	External signals	Reporting / experimental detail
Sierla et al. (2022)	3	1	1	0	2	3

Table S6.27: RQ6 critical-audit scores for Sierla et al. (2022), recoded using the shared evidence-intensity scale.

Survey	Benchmark comparability critique	Open-code / reproducibility critique	Simulator fragmentation critique	Data realism / availability critique	Standardization / shared-infrastructure need
Sierla et al. (2022)	3	0	3	1	3

Table S6.28: RQ6 recurring infrastructure-gap scores for Sierla et al. (2022), recoded using the shared evidence-intensity scale.

Survey	Closed code / unavailable artifacts	Lack of shared benchmarks	Inconsistent external signals / assumptions	Simulator / testbed fragmentation	Weak dataset realism / documentation
Sierla et al. (2022)	0	2	2	3	1

Table S6.29: RQ6 summary table for Sierla et al. (2022), after recoding.

Survey	Reproducibility orientation	Dominant infrastructure concern	Main RQ6 consequence
Sierla et al. (2022)	Comparability- and reporting-oriented, with a strong environment-realism critique	The survey’s dominant infrastructure concern is that studies differ too much in abstraction level, action-space design, simulator realism, virtual-versus-physical environment choice, and reporting clarity for results to be compared cleanly or interfaced confidently with real HVAC systems.	The survey contributes strongly to RQ6 by showing that reproducibility problems in RL-for-HVAC are not only about open artifacts, but also about whether environments, action formulations, abstraction levels, and reporting detail are specified clearly enough to support comparison and later deployment.

Table S6.30: One-line RQ6 answer for Sierla et al. (2022), after recoding.

Survey	RQ6 answer
Sierla et al. (2022)	Sierla et al. (2022) answer RQ6 mainly by showing that RL-for-HVAC studies are often hard to compare and reproduce at the control-design level because simulator realism, action formulations, abstraction levels, external assumptions, and reporting detail vary substantially across the literature, even when reward-focused results look promising.

S6.6 Applications of reinforcement learning for building energy efficiency control: A review [6]

Table S6.31: RQ6 recoded audit table for Fu et al. (2022). Scores use the shared 0–3 evidence-intensity scale.

Survey	Reproducibility auditing	Infrastructure auditing	Blind-spot visibility	Recoded evidence justification
Fu et al. (2022)	1	2	1	The review is broadly infrastructure-aware but only lightly reproducibility-critical. Reproducibility auditing receives score 1 because the paper does not systematically evaluate open code, shared benchmark tasks, artifact availability, or dataset openness. Infrastructure auditing receives score 2 because the paper discusses simulation platforms as a practical response to the risks and costs of real-building training, including examples such as BCVTB with EnergyPlus and CitySim with TensorFlow. Blind-spot visibility receives score 1 because the paper recognizes simulator dependence, training time, environment modeling, and real-building training risk, but does not develop these into a strong reproducibility or shared-infrastructure critique.

Table S6.32: RQ6 infrastructure-coverage scores for Fu et al. (2022), recoded using the shared evidence-intensity scale.

Survey	Simulators / environments	Datasets / data sources	Benchmark tasks	Code / artifact availability	External signals	Reporting / experimental detail
Fu et al. (2022)	2	0	0	0	1	2

Table S6.33: RQ6 critical-audit scores for Fu et al. (2022), recoded using the shared evidence-intensity scale.

Survey	Benchmark comparability critique	Open-code / reproducibility critique	Simulator fragmentation critique	Data realism / availability critique	Standardization / shared-infrastructure need
Fu et al. (2022)	0	0	1	0	0

Table S6.34: RQ6 recurring infrastructure-gap scores for Fu et al. (2022), recoded using the shared evidence-intensity scale.

Survey	Closed code / unavailable artifacts	Lack of shared benchmarks	Inconsistent external signals / assumptions	Simulator / testbed fragmentation	Weak dataset realism / documentation
Fu et al. (2022)	0	0	1	1	0

Table S6.35: RQ6 summary table for Fu et al. (2022), after recoding.

Survey	Reproducibility orientation	Dominant infrastructure concern	Main RQ6 consequence
Fu et al. (2022)	Broadly infrastructure-aware, but only lightly reproducibility-critical	The survey’s dominant infrastructure concern is that reinforcement learning for building energy efficiency depends heavily on simulation platforms because real-building training is risky, costly, and time-consuming.	The survey contributes to RQ6 mainly by showing that simulator dependence and training burden are central practical bottlenecks, but it remains weak on open-artifact availability, shared benchmarks, and dataset-centered reproducibility critique.

Table S6.36: One-line RQ6 answer for Fu et al. (2022), after recoding.

Survey	RQ6 answer
Fu et al. (2022)	Fu et al. (2022) answer RQ6 mainly by showing that reinforcement-learning-based building-energy-efficiency control still relies strongly on simulation environments because direct real-building training is risky and costly, while giving only limited attention to shared datasets, benchmark infrastructure, open code, or artifact reproducibility.

S6.7 Systematic review on deep reinforcement learning-based energy management for different building types [7]

Table S6.37: RQ6 recoded audit table for Shaqour and Hagishima (2022). Scores use the shared 0–3 evidence-intensity scale.

Survey	Reproducibility auditing	Infrastructure auditing	Blind-spot visibility	Recoded evidence justification
Shaqour and Hagishima (2022)	3	3	3	The review is strongly infrastructure- and reproducibility-aware. Reproducibility auditing receives score 3 because the paper includes explicit discussion of reproducibility, open-sourcing code, standardized open datasets, and the need to improve comparison and reuse. Infrastructure auditing receives score 3 because the paper discusses CityLearn as an open-source environment intended to support standardized comparison of reinforcement-learning algorithms for building energy management. Blind-spot visibility receives score 3 because the paper links data hunger, limited real-world validation, lack of openness, and insufficient standardization to weak comparability and reproducibility in DRL-based BEMS research.

Table S6.38: RQ6 infrastructure-coverage scores for Shaqour and Hagishima (2022), recoded using the shared evidence-intensity scale.

Survey	Simulators / environments	Datasets / data sources	Benchmark tasks	Code / artifact availability	External signals	Reporting / experimental detail
Shaqour and Hagishima (2022)	3	3	3	3	2	3

Table S6.39: RQ6 critical-audit scores for Shaqour and Hagishima (2022), recoded using the shared evidence-intensity scale.

Survey	Benchmark comparability critique	Open-code / reproducibility critique	Simulator fragmentation critique	Data realism / availability critique	Standardization / shared-infrastructure need
Shaqour and Hagishima (2022)	3	3	1	3	3

Table S6.40: RQ6 recurring infrastructure-gap scores for Shaqour and Hagishima (2022), recoded using the shared evidence-intensity scale.

Survey	Closed code / unavailable artifacts	Lack of shared benchmarks	Inconsistent external signals / assumptions	Simulator / testbed fragmentation	Weak dataset realism / documentation
Shaqour and Hagishima (2022)	3	3	1	1	3

Table S6.41: RQ6 summary table for Shaqour and Hagishima (2022), after recoding.

Survey	Reproducibility orientation	Dominant infrastructure concern	Main RQ6 consequence
Shaqour and Hagishima (2022)	Strongly infrastructure- and reproducibility-aware, especially for an early survey	The survey’s dominant infrastructure concern is that DRL-based BEMS remains constrained by large data requirements, scarce real validation, and insufficiently standardized open benchmarking and sharing practices.	The survey contributes strongly to RQ6 by combining systematic cross-building-type mapping with an explicit call for open-source environments, open code, and standardized datasets, showing that infrastructure weakness in the field is not only about scarce deployment but also about limited comparability and reuse.

Table S6.42: One-line RQ6 answer for Shaqour and Hagishima (2022), after recoding.

Survey	RQ6 answer
Shaqour and Hagishima (2022)	Shaqour and Hagishima (2022) answer RQ6 by showing that DRL-based BEMS research is still constrained by data hunger and weak real-world validation, while also arguing that stronger reproducibility and comparability require open-source environments such as CityLearn, open code, and more standardized open datasets.

S6.8 Building energy management with reinforcement learning and model predictive control: A survey [8]

Table S6.43: RQ6 recoded audit table for Zhang et al. (2022). Scores use the shared 0–3 evidence-intensity scale.

Survey	Reproducibility auditing	Infrastructure auditing	Blind-spot visibility	Recoded evidence justification
Zhang et al. (2022)	1	2	2	The review is control- and modeling-infrastructure aware, but only lightly reproducibility-critical. Reproducibility auditing receives score 1 because the paper does not systematically audit open code, shared benchmark tasks, artifact availability, or reproducible experimental packages. Infrastructure auditing receives score 2 because the paper discusses data-driven MPC and RL for BEMS, historical operational data, MPC-generated training data, accurate simulators, learned environment models, forecasts, future disturbances, occupancy, human interaction, and external uncertainty. Blind-spot visibility receives score 2 because the paper identifies data efficiency, safety, robust adaptability, simulator cost, and learning from historical data as practical limitations, but does not develop a strong shared-benchmark or open-artifact reproducibility agenda.

Table S6.44: RQ6 infrastructure-coverage scores for Zhang et al. (2022), recoded using the shared evidence-intensity scale.

Survey	Simulators / environments	Datasets / data sources	Benchmark tasks	Code / artifact availability	External signals	Reporting / experimental detail
Zhang et al. (2022)	2	2	0	0	2	2

Table S6.45: RQ6 critical-audit scores for Zhang et al. (2022), recoded using the shared evidence-intensity scale.

Survey	Benchmark comparability critique	Open-code / reproducibility critique	Simulator fragmentation critique	Data realism / availability critique	Standardization / shared-infrastructure need
Zhang et al. (2022)	0	0	1	2	0

Table S6.46: RQ6 recurring infrastructure-gap scores for Zhang et al. (2022), recoded using the shared evidence-intensity scale.

Survey	Closed code / unavailable artifacts	Lack of shared benchmarks	Inconsistent external signals / assumptions	Simulator / testbed fragmentation	Weak dataset realism / documentation
Zhang et al. (2022)	0	0	1	1	1

Table S6.47: RQ6 summary table for Zhang et al. (2022), after recoding.

Survey	Reproducibility orientation	Dominant infrastructure concern	Main RQ6 consequence
Zhang et al. (2022)	Control- and modeling-infrastructure aware, but only lightly reproducibility-critical	The survey’s dominant infrastructure concern is that practical BEMS control depends on data sources, forecast information, simulators, learned environment models, and historically collected operational data, while data efficiency and safety remain unresolved.	The survey contributes to RQ6 mainly by making the data, model, simulator, and forecast infrastructure behind MPC- and RL-based BEMS more explicit, but it remains weaker on shared benchmarks, open artifacts, and direct reproducibility critique than the more infrastructure-auditing surveys.

Table S6.48: One-line RQ6 answer for Zhang et al. (2022), after recoding.

Survey	RQ6 answer
Zhang et al. (2022)	Zhang et al. (2022) answer RQ6 mainly by showing that BEMS control research depends on nontrivial data, model, simulator, and forecast infrastructure, while practical RL remains limited by data efficiency, safety, robust adaptability, and the cost of accurate environment modeling rather than by an explicitly developed benchmark or open-artifact culture.

S6.9 Reinforcement learning: theory and applications in HEMS [9]

Table S6.49: RQ6 recoded audit table for Al-Ani and Das (2022). Scores use the shared 0–3 evidence-intensity scale.

Survey	Reproducibility auditing	Infrastructure auditing	Blind-spot visibility	Recoded evidence justification
Al-Ani and Das (2022)	1	2	1	The review is broadly infrastructure-aware but weakly reproducibility-critical. Reproducibility auditing receives score 1 because the paper does not systematically audit open code, shared datasets, benchmark environments, artifact availability, or standardization needs across the literature. Infrastructure auditing receives score 2 because the paper discusses HEMS enabling technologies and infrastructure, including communication protocols, IoT-enabled devices, wireless communication, HAN/WAN structures, interoperability, scalability, simulation/model-based training, and limited actual HEMS deployment. Blind-spot visibility receives score 1 because the paper signals that RL in HEMS remains early-stage and largely simulation-based, but does not develop this into a strong reproducibility, open-artifact, or shared-benchmark critique.

Table S6.50: RQ6 infrastructure-coverage scores for Al-Ani and Das (2022), recoded using the shared evidence-intensity scale.

Survey	Simulators / environments	Datasets / data sources	Benchmark tasks	Code / artifact availability	External signals	Reporting / experimental detail
Al-Ani and Das (2022)	2	0	0	0	1	2

Table S6.51: RQ6 critical-audit scores for Al-Ani and Das (2022), recoded using the shared evidence-intensity scale.

Survey	Benchmark comparability critique	Open-code / reproducibility critique	Simulator fragmentation critique	Data realism / availability critique	Standardization / shared-infrastructure need
Al-Ani and Das (2022)	0	0	0	1	0

Table S6.52: RQ6 recurring infrastructure-gap scores for Al-Ani and Das (2022), recoded using the shared evidence-intensity scale.

Survey	Closed code / unavailable artifacts	Lack of shared benchmarks	Inconsistent external signals / assumptions	Simulator / testbed fragmentation	Weak dataset realism / documentation
Al-Ani and Das (2022)	0	0	0	0	1

Table S6.53: RQ6 summary table for Al-Ani and Das (2022), after recoding.

Survey	Reproducibility orientation	Dominant infrastructure concern	Main RQ6 consequence
Al-Ani and Das (2022)	Broadly infrastructure-aware, but weakly reproducibility-critical	The survey’s dominant infrastructure concern is that HEMS reinforcement learning remains early-stage and depends on enabling communication, IoT, and home-network infrastructure, while still relying largely on model-based or simulation-based training rather than mature real-world experimentation.	The survey contributes to RQ6 mainly by signaling the immaturity and simulation dependence of the HEMS reinforcement-learning literature, but it remains weak on datasets, shared benchmarks, open artifacts, and direct reproducibility critique.

Table S6.54: One-line RQ6 answer for Al-Ani and Das (2022), after recoding.

Survey	RQ6 answer
Al-Ani and Das (2022)	Al-Ani and Das (2022) answer RQ6 mainly by showing that reinforcement learning in HEMS is still early-stage and largely simulation-based, while depending on communication, IoT, and home-network infrastructure, and giving only limited attention to datasets, benchmark environments, shared artifacts, and reproducibility infrastructure needed for stronger cross-study comparability.

S6.10 A review of reinforcement learning for controlling building energy systems from a computer science perspective [10]

Table S6.55: RQ6 recoded audit table for Weinberg et al. (2023). Scores use the shared 0–3 evidence-intensity scale.

Survey	Reproducibility auditing	Infrastructure auditing	Blind-spot visibility	Recoded evidence justification
Weinberg et al. (2023)	2	3	3	The review is system- and infrastructure-aware, but only moderately reproducibility-critical. Reproducibility auditing receives score 2 because the paper does not audit open artifacts or common benchmark datasets, but it structurally discusses sample efficiency, slow data collection, transfer difficulty, and data-processing conditions that affect comparability and practical reproducibility. Infrastructure auditing receives score 3 because the paper explicitly frames practical RL for building energy systems around large-scale data collection, BMS sensing and actuation, scalable computation pipelines, cloud-based processing, communication between sensors and actuators, interoperability, and asynchronous multi-building data streams. Blind-spot visibility receives score 3 because sample-efficiency constraints, heterogeneous building dynamics, transfer difficulty, interoperability barriers, and scalable system design are structural computer-science challenges and future directions in the review.

Table S6.56: RQ6 infrastructure-coverage scores for Weinberg et al. (2023), recoded using the shared evidence-intensity scale.

Survey	Simulators / environments	Datasets / data sources	Benchmark tasks	Code / artifact availability	External signals	Reporting / experimental detail
Weinberg et al. (2023)	2	3	0	0	2	2

Table S6.57: RQ6 critical-audit scores for Weinberg et al. (2023), recoded using the shared evidence-intensity scale.

Survey	Benchmark comparability critique	Open-code / reproducibility critique	Simulator fragmentation critique	Data realism / availability critique	Standardization / shared-infrastructure need
Weinberg et al. (2023)	0	0	1	3	3

Table S6.58: RQ6 recurring infrastructure-gap scores for Weinberg et al. (2023), recoded using the shared evidence-intensity scale.

Survey	Closed code / unavailable artifacts	Lack of shared benchmarks	Inconsistent external signals / assumptions	Simulator / testbed fragmentation	Weak dataset realism / documentation
Weinberg et al. (2023)	0	0	1	2	3

Table S6.59: RQ6 summary table for Weinberg et al. (2023), after recoding.

Survey	Reproducibility orientation	Dominant infrastructure concern	Main RQ6 consequence
Weinberg et al. (2023)	System- and infrastructure-aware, but only moderately reproducibility-critical	The survey’s dominant infrastructure concern is that practical reinforcement learning for building energy systems depends on slow and costly data collection, transferable environment knowledge, scalable processing, communication infrastructure, interoperability, and asynchronous multi-building data streams.	The survey contributes to RQ6 mainly by showing that reproducibility and comparability in reinforcement learning for building energy systems are constrained by sample efficiency, transfer difficulty, heterogeneous building dynamics, asynchronous data pipelines, and interoperability challenges, even though the paper is less focused on open artifacts or shared benchmark ecosystems.

Table S6.60: One-line RQ6 answer for Weinberg et al. (2023), after recoding.

Survey	RQ6 answer
Weinberg et al. (2023)	Weinberg et al. (2023) answer RQ6 mainly by showing that reinforcement learning for building energy systems relies on demanding data-collection, transfer, communication, and computation infrastructure, and that practical progress is limited less by missing algorithm ideas than by sample-efficiency constraints, heterogeneous building dynamics, and the difficulty of scaling interoperable control pipelines.

S6.11 Dynamic indoor thermal environment using reinforcement learning-based controls: Opportunities and challenges [11]

Table S6.61: RQ6 recoded audit table for Chatterjee and Khovalyg (2023). Scores use the shared 0–3 evidence-intensity scale.

Survey	Reproducibility auditing	Infrastructure auditing	Blind-spot visibility	Recoded evidence justification
Chatterjee and Khovalyg (2023)	3	3	3	The review is strongly simulator-, reporting-, and benchmarking-aware. Reproducibility auditing receives score 3 because the paper uses explicit inclusion criteria, records building type, HVAC system, DRL algorithm, simulation framework, controlled parameters, and implementation status, and makes these reporting dimensions part of its synthesis. Infrastructure auditing receives score 3 because simulation software, building model scale, building type, virtual environments, and real-building implementation status are structurally visible in the review. Blind-spot visibility receives score 3 because the paper explicitly identifies the lack of a data-rich, open-source, interoperable virtual testbed as a barrier to cross-study validation and benchmarking.

Table S6.62: RQ6 infrastructure-coverage scores for Chatterjee and Khovalyg (2023), recoded using the shared evidence-intensity scale.

Survey	Simulators / environments	Datasets / data sources	Benchmark tasks	Code / artifact availability	External signals	Reporting / experimental detail
Chatterjee and Khovalyg (2023)	3	2	3	1	2	3

Table S6.63: RQ6 critical-audit scores for Chatterjee and Khovalyg (2023), recoded using the shared evidence-intensity scale.

Survey	Benchmark comparability critique	Open-code / reproducibility critique	Simulator fragmentation critique	Data realism / availability critique	Standardization / shared-infrastructure need
Chatterjee and Khovalyg (2023)	3	1	3	2	3

Table S6.64: RQ6 recurring infrastructure-gap scores for Chatterjee and Khovalyg (2023), recoded using the shared evidence-intensity scale.

Survey	Closed code / unavailable artifacts	Lack of shared benchmarks	Inconsistent external signals / assumptions	Simulator / testbed fragmentation	Weak dataset realism / documentation
Chatterjee and Khovalyg (2023)	1	3	1	3	2

Table S6.65: RQ6 summary table for Chatterjee and Khovalyg (2023), after recoding.

Survey	Reproducibility orientation	Dominant infrastructure concern	Main RQ6 consequence
Chatterjee and Khovalyg (2023)	Strongly simulator- and benchmarking-aware, with moderate open-artifact critique	The survey’s dominant infrastructure concern is that RL-based HVAC research remains heavily simulation-dependent and lacks sufficiently rich, shared, open, and interoperable virtual infrastructure for fair cross-study validation and benchmarking.	The survey contributes strongly to RQ6 by showing that comparability in RL-HVAC research depends on clear reporting of simulation environments, control settings, building types, and implementation status, while broader progress requires more standardized, open, and interoperable virtual testbeds rather than isolated simulation studies alone.

Table S6.66: One-line RQ6 answer for Chatterjee and Khovalyg (2023), after recoding.

Survey	RQ6 answer
Chatterjee and Khovalyg (2023)	Chatterjee and Khovalyg (2023) answer RQ6 by showing that RL-based HVAC research is still dominated by simulation environments, only partly validated in real buildings, and in need of a data-rich, open-source, interoperable virtual testbed to enable clearer cross-study validation, benchmarking, and more comparable infrastructure practices.

S6.12 Reinforcement learning for HVAC control in intelligent buildings: A technical and conceptual review [12]

Table S6.67: RQ6 recoded audit table for Al Sayed et al. (2024). Scores use the shared 0–3 evidence-intensity scale.

Survey	Reproducibility auditing	Infrastructure auditing	Blind-spot visibility	Recoded evidence justification
Al Sayed et al. (2024)	3	3	3	The review is strongly environment- and data-realism-aware. Reproducibility auditing receives score 3 because the paper structurally audits training settings, simulator/model categories, exogenous-state sequences, and the generalization consequences of repeated or insufficiently diverse training conditions. Infrastructure auditing receives score 3 because the review gives detailed attention to training environments and simulation platforms, including EnergyPlus, TRNSYS, Dymola, BOPTEST, IESVE, CityLearn, MATLAB, and TESP, and distinguishes white-box, grey-box, and black-box approaches. Blind-spot visibility receives score 3 because weak diversification of exogenous state components such as occupancy schedules, electricity prices, and weather is treated as a central blind spot for reproducibility, generalization, and deployment.

Table S6.68: RQ6 infrastructure-coverage scores for Al Sayed et al. (2024), recoded using the shared evidence-intensity scale.

Survey	Simulators / environments	Datasets / data sources	Benchmark tasks	Code / artifact availability	External signals	Reporting / experimental detail
Al Sayed et al. (2024)	3	3	2	0	3	3

Table S6.69: RQ6 critical-audit scores for Al Sayed et al. (2024), recoded using the shared evidence-intensity scale.

Survey	Benchmark comparability critique	Open-code / reproducibility critique	Simulator fragmentation critique	Data realism / availability critique	Standardization / shared-infrastructure need
Al Sayed et al. (2024)	2	0	3	3	3

Table S6.70: RQ6 recurring infrastructure-gap scores for Al Sayed et al. (2024), recoded using the shared evidence-intensity scale.

Survey	Closed code / unavailable artifacts	Lack of shared benchmarks	Inconsistent external signals / assumptions	Simulator / testbed fragmentation	Weak dataset realism / documentation
Al Sayed et al. (2024)	0	2	3	3	3

Table S6.71: RQ6 summary table for Al Sayed et al. (2024), after recoding.

Survey	Reproducibility orientation	Dominant infrastructure concern	Main RQ6 consequence
Al Sayed et al. (2024)	Strongly environment- and data-realism-aware, with moderate open-artifact critique	The survey’s dominant infrastructure concern is that RL-HVAC training environments often fail to represent exogenous variability realistically enough, especially weather, occupancy, and electricity-price variation, which weakens transfer from simulation to real deployment.	The survey contributes strongly to RQ6 by showing that reproducibility and deployment readiness depend heavily on simulator choice, exogenous-state design, and real-world data realism, even though the paper is less focused on open artifacts or code-sharing practices.

Table S6.72: One-line RQ6 answer for Al Sayed et al. (2024), after recoding.

Survey	RQ6 answer
Al Sayed et al. (2024)	Al Sayed et al. (2024) answer RQ6 mainly by showing that RL-for-HVAC research depends strongly on simulator realism, exogenous-state diversity, and real-world data support, and that weak handling of weather, occupancy, and price variability can undermine both reproducibility and later deployment.

S6.13 An overview of reinforcement learning-based approaches for smart home energy management systems with energy storages [13]

Table S6.73: RQ6 recoded audit table for Pinthurat et al. (2024). Scores use the shared 0–3 evidence-intensity scale.

Survey	Reproducibility auditing	Infrastructure auditing	Blind-spot visibility	Recoded evidence justification
Pinthurat et al. (2024)	1	2	2	The review is broadly infrastructure-aware but only lightly reproducibility-critical. Reproducibility auditing receives score 1 because the paper does not systematically audit open code, benchmark environments, dataset sharing, artifact availability, or reproducible experimental packages. Infrastructure auditing receives score 2 because the paper frames RL-based smart home energy management with storage as depending on controllers, sensors, actuators, connectivity, software platforms, smart meters, grid interaction, renewable integration, energy storage, pricing signals, weather conditions, user preferences, demand-response signals, and grid state. Blind-spot visibility receives score 2 because the paper identifies sample inefficiency, data intensity, real-world testing needs, and sensitivity to actual user behavior, weather, and grid dynamics as practical limitations, although not as a strong open-benchmark or open-artifact agenda.

Table S6.74: RQ6 infrastructure-coverage scores for Pinthurat et al. (2024), recoded using the shared evidence-intensity scale.

Survey	Simulators / environments	Datasets / data sources	Benchmark tasks	Code / artifact availability	External signals	Reporting / experimental detail
Pinthurat et al. (2024)	1	2	0	0	3	2

Table S6.75: RQ6 critical-audit scores for Pinthurat et al. (2024), recoded using the shared evidence-intensity scale.

Survey	Benchmark comparability critique	Open-code / reproducibility critique	Simulator fragmentation critique	Data realism / availability critique	Standardization / shared-infrastructure need
Pinthurat et al. (2024)	0	0	0	2	0

Table S6.76: RQ6 recurring infrastructure-gap scores for Pinthurat et al. (2024), recoded using the shared evidence-intensity scale.

Survey	Closed code / unavailable artifacts	Lack of shared benchmarks	Inconsistent external signals / assumptions	Simulator / testbed fragmentation	Weak dataset realism / documentation
Pinthurat et al. (2024)	0	0	2	0	2

Table S6.77: RQ6 summary table for Pinthurat et al. (2024), after recoding.

Survey	Reproducibility orientation	Dominant infrastructure concern	Main RQ6 consequence
Pinthurat et al. (2024)	Broadly infrastructure-aware, but only lightly reproducibility-critical	The survey’s dominant infrastructure concern is that RL-based smart-home energy management with storage depends heavily on external signals and broader smart-home system components, while real-world validation under changing user, weather, and grid conditions remains limited.	The survey contributes to RQ6 mainly by making exogenous-signal dependence and smart-home system infrastructure explicit, but it remains weak on benchmark environments, open artifacts, and stronger reproducibility critique.

Table S6.78: One-line RQ6 answer for Pinthurat et al. (2024), after recoding.

Survey	RQ6 answer
Pinthurat et al. (2024)	Pinthurat et al. (2024) answer RQ6 mainly by showing that RL-based smart-home energy management with storage depends on a broad control and data infrastructure and is especially sensitive to exogenous variables such as pricing, weather, user behavior, and grid dynamics, while giving only limited attention to shared benchmarks, open code, or artifact reproducibility.

S6.14 Applications of Deep Reinforcement Learning for Home Energy Management Systems: A Review [14]

Table S6.79: RQ6 recoded audit table for Latoń et al. (2024). Scores use the shared 0–3 evidence-intensity scale.

Survey	Reproducibility auditing	Infrastructure auditing	Blind-spot visibility	Recoded evidence justification
Latoń et al. (2024)	2	3	3	The review is strongly infrastructure-aware, with moderate reproducibility critique. Reproducibility auditing receives score 2 because the paper discusses data frameworks, privacy/security, interoperability, digital twins, and real-world case-study gaps, but gives little direct attention to open code or artifact availability. Infrastructure auditing receives score 3 because the paper structurally frames RL/DRL-based HEMS as depending on IoT devices, distributed control networks, smart-grid interaction, cloud and edge computing, data-collection pipelines, communication protocols, privacy-preserving mechanisms, digital twins, and scalable deployment infrastructure. Blind-spot visibility receives score 3 because limited real-world case studies, noisy and incomplete data, interoperability challenges, scalability limits, privacy/security risks, and insufficiently mature digital-twin infrastructure are explicit problems and future needs.

Table S6.80: RQ6 infrastructure-coverage scores for Latoń et al. (2024), recoded using the shared evidence-intensity scale.

Survey	Simulators / environments	Datasets / data sources	Benchmark tasks	Code / artifact availability	External signals	Reporting / experimental detail
Latoń et al. (2024)	3	3	2	0	3	3

Table S6.81: RQ6 critical-audit scores for Latoń et al. (2024), recoded using the shared evidence-intensity scale.

Survey	Benchmark comparability critique	Open-code / reproducibility critique	Simulator fragmentation critique	Data realism / availability critique	Standardization / shared-infrastructure need
Latoń et al. (2024)	2	0	1	3	3

Table S6.82: RQ6 recurring infrastructure-gap scores for Latoń et al. (2024), recoded using the shared evidence-intensity scale.

Survey	Closed code / unavailable artifacts	Lack of shared benchmarks	Inconsistent external signals / assumptions	Simulator / testbed fragmentation	Weak dataset realism / documentation
Latoń et al. (2024)	0	2	2	2	3

Table S6.83: RQ6 summary table for Latoń et al. (2024), after recoding.

Survey	Reproducibility orientation	Dominant infrastructure concern	Main RQ6 consequence
Latoń et al. (2024)	Strongly infrastructure-aware, with moderate reproducibility critique	The survey’s dominant infrastructure concern is that RL/DRL-based HEMS depends on robust data pipelines, interoperable communication infrastructure, privacy and security mechanisms, scalable deployment infrastructure, and realistic digital-twin environments, while real-world case evidence remains limited.	The survey contributes strongly to RQ6 by showing that reproducibility and practical comparability in HEMS are constrained not only by simulation-heavy evaluation, but also by weak data realism, interoperability challenges, scalability limits, privacy/security requirements, and insufficiently mature real-world infrastructure.

Table S6.84: One-line RQ6 answer for Latoń et al. (2024), after recoding.

Survey	RQ6 answer
Latoń et al. (2024)	Latoń et al. (2024) answer RQ6 mainly by showing that RL/DRL-based HEMS depends on substantial data, communication, privacy, interoperability, scalability, and digital-twin infrastructure, and that stronger benchmarking and real-world case evidence are still needed before the field can claim mature and broadly reusable research infrastructure.

S6.15 Multi objectives reinforcement learning for smart buildings: A systematic review of algorithms, applications and future perspectives [15]

Table S6.85: RQ6 recoded audit table for Huynh et al. (2025). Scores use the shared 0–3 evidence-intensity scale.

Survey	Reproducibility auditing	Infrastructure auditing	Blind-spot visibility	Recoded evidence justification
Huynh et al. (2025)	3	3	3	The review is strongly data- and reproducibility-aware, with explicit attention to open-source practices. Reproducibility auditing receives score 3 because the paper explicitly includes open data and code-sharing practices as part of its contribution and records open-source implementations, programming language, simulation environment, historical-data use, weather and pricing inputs, and reproducibility-supporting tools such as OpenAI Gym and Docker. Infrastructure auditing receives score 3 because the paper makes simulation/testbed infrastructure, historical data, real-world input streams, programming platforms, and open-source implementation support structurally visible. Blind-spot visibility receives score 3 because data quality, noisy or incomplete sensor and communication data, computational complexity, heterogeneity across building types and HVAC configurations, scalability, interpretability, transparency, and stakeholder trust are recurring barriers to reusable and deployable MORL infrastructure.

Table S6.86: RQ6 infrastructure-coverage scores for Huynh et al. (2025), recoded using the shared evidence-intensity scale.

Survey	Simulators / environments	Datasets / data sources	Benchmark tasks	Code / artifact availability	External signals	Reporting / experimental detail
Huynh et al. (2025)	3	3	2	3	3	3

Table S6.87: RQ6 critical-audit scores for Huynh et al. (2025), recoded using the shared evidence-intensity scale.

Survey	Benchmark comparability critique	Open-code / reproducibility critique	Simulator fragmentation critique	Data realism / availability critique	Standardization / shared-infrastructure need
Huynh et al. (2025)	2	3	2	3	3

Table S6.88: RQ6 recurring infrastructure-gap scores for Huynh et al. (2025), recoded using the shared evidence-intensity scale.

Survey	Closed code / unavailable artifacts	Lack of shared benchmarks	Inconsistent external signals / assumptions	Simulator / testbed fragmentation	Weak dataset realism / documentation
Huynh et al. (2025)	2	2	2	2	3

Table S6.89: RQ6 summary table for Huynh et al. (2025), after recoding.			
Survey	Reproducibility orientation	Dominant infrastructure concern	Main RQ6 consequence
Huynh et al. (2025)	Strongly data- and reproducibility-aware, with explicit attention to open-source practices	The survey’s dominant infrastructure concern is that MORL for smart buildings depends on high-quality data, open and reusable implementation support, simulation/testbed infrastructure, and methods that can handle heterogeneous buildings, external signals, noisy real-world data streams, and computational scalability.	The survey contributes strongly to RQ6 by making code-sharing practices, simulation/testbed support, real-world data inputs, and infrastructure heterogeneity visible, while also showing that stronger reproducibility still depends on better data quality, broader reuse, and more adaptable methods across diverse building settings.

Table S6.90: One-line RQ6 answer for Huynh et al. (2025), after recoding.	
Survey	RQ6 answer
Huynh et al. (2025)	Huynh et al. (2025) answer RQ6 by showing that MORL for smart buildings relies on substantial data, simulation/testbed, and open-source implementation infrastructure, while practical reproducibility and reuse are still constrained by noisy or incomplete data, heterogeneous building contexts, computational complexity, and limited standardization across studies.

S6.16 Reinforcement learning for control and optimization of real buildings: Identifying and addressing implementation hurdles [16]

Table S6.91: RQ6 recoded audit table for Kannari et al. (2025). Scores use the shared 0–3 evidence-intensity scale.

Survey	Reproducibility auditing	Infrastructure auditing	Blind-spot visibility	Recoded evidence justification
Kannari et al. (2025)	3	3	3	The review is strongly implementation- and validation-infrastructure aware, with a high level of reproducibility critique. Reproducibility auditing receives score 3 because the paper discusses comparable validation protocols, short and heterogeneous test periods, black-box market solutions, changing weather and occupancy, safety checks, fallback logic, governance, transparency, accountability, and the need for agreed digital-twin validation criteria. Infrastructure auditing receives score 3 because real-building RL is framed as depending on virtual pretraining, online deployment design, relearning loops, data preprocessing, recalibration, safety checks, digital-twin maturity, data sharing, security mechanisms, and IoT maturity. Blind-spot visibility receives score 3 because weak standardization of digital twins, limited data sharing, short test periods, black-box implementation practices, privacy risks, and nonstandard validation criteria are explicit limitations for reproducible real-building RL.

Table S6.92: RQ6 infrastructure-coverage scores for Kannari et al. (2025), recoded using the shared evidence-intensity scale.

Survey	Simulators / environments	Datasets / data sources	Benchmark tasks	Code / artifact availability	External signals	Reporting / experimental detail
Kannari et al. (2025)	3	3	2	1	2	3

Table S6.93: RQ6 critical-audit scores for Kannari et al. (2025), recoded using the shared evidence-intensity scale.

Survey	Benchmark comparability critique	Open-code / reproducibility critique	Simulator fragmentation critique	Data realism / availability critique	Standardization / shared-infrastructure need
Kannari et al. (2025)	3	1	2	3	3

Table S6.94: RQ6 recurring infrastructure-gap scores for Kannari et al. (2025), recoded using the shared evidence-intensity scale.

Survey	Closed code / unavailable artifacts	Lack of shared benchmarks	Inconsistent external signals / assumptions	Simulator / testbed fragmentation	Weak dataset realism / documentation
Kannari et al. (2025)	2	2	2	2	3

Table S6.95: RQ6 summary table for Kannari et al. (2025), after recoding.

Survey	Reproducibility orientation	Dominant infrastructure concern	Main RQ6 consequence
Kannari et al. (2025)	Strongly implementation- and validation-infrastructure aware, with a high level of reproducibility critique	The survey’s dominant infrastructure concern is that real-building reinforcement learning still lacks sufficiently mature and standardized pretraining, deployment, validation, digital-twin, data-sharing, and governance infrastructure for reliable cross-case comparison and wider adoption.	The survey contributes strongly to RQ6 by showing that reproducibility in real-building reinforcement learning depends not only on algorithms, but also on comparable validation protocols, data quality management, digital-twin maturity, security and privacy mechanisms, transparency, governance, and implementation practices.

Table S6.96: One-line RQ6 answer for Kannari et al. (2025), after recoding.

Survey	RQ6 answer
Kannari et al. (2025)	Kannari et al. (2025) answer RQ6 by showing that real-building reinforcement learning depends on robust pretraining, online deployment, validation, digital-twin, data-sharing, and governance infrastructure, and that reproducibility remains limited by short test periods, changing weather and occupancy conditions, nonstandardized validation criteria, limited transparency, and weak data-sharing practices.

S6.17 Reinforcement learning for optimizing renewable energy utilization in buildings: A review on applications and innovations [17]

Table S6.97: RQ6 recoded audit table for Michailidis et al. (2025). Scores use the shared 0–3 evidence-intensity scale.

Survey	Reproducibility auditing	Infrastructure auditing	Blind-spot visibility	Recoded evidence justification
Michailidis et al. (2025)	1	2	1	The review is structured and implementation-aware, but only lightly reproducibility-critical. Reproducibility auditing receives score 1 because the paper uses structured literature analysis and summary tables, but gives limited direct attention to open-source code, shared datasets, benchmark ecosystems, or artifact reproducibility. Infrastructure auditing receives score 2 because the paper makes implementation attributes and external-signal assumptions visible through dimensions such as RL methodology, agent framework, target outputs, baseline control strategies, RES configurations, BEMS types, building typologies, electricity prices, demand-response requests, weather, occupancy, renewable generation, and battery state of charge. Blind-spot visibility receives score 1 because some implementation and external-signal assumptions are visible, but reproducibility gaps are not a major organizing critique.

Table S6.98: RQ6 infrastructure-coverage scores for Michailidis et al. (2025), recoded using the shared evidence-intensity scale.

Survey	Simulators / environments	Datasets / data sources	Benchmark tasks	Code / artifact availability	External signals	Reporting / experimental detail
Michailidis et al. (2025)	2	1	0	0	3	3

Table S6.99: RQ6 critical-audit scores for Michailidis et al. (2025), recoded using the shared evidence-intensity scale.

Survey	Benchmark comparability critique	Open-code / reproducibility critique	Simulator fragmentation critique	Data realism / availability critique	Standardization / shared-infrastructure need
Michailidis et al. (2025)	1	0	0	1	1

Table S6.100: RQ6 recurring infrastructure-gap scores for Michailidis et al. (2025), recoded using the shared evidence-intensity scale.

Survey	Closed code / unavailable artifacts	Lack of shared benchmarks	Inconsistent external signals / assumptions	Simulator / testbed fragmentation	Weak dataset realism / documentation
Michailidis et al. (2025)	0	0	1	0	1

Table S6.101: RQ6 summary table for Michailidis et al. (2025), after recoding.

Survey	Reproducibility orientation	Dominant infrastructure concern	Main RQ6 consequence
Michailidis et al. (2025)	Structured and implementation-aware, but only lightly reproducibility-critical	The survey’s dominant infrastructure concern is that RL-based renewable-energy building control depends on clearly specified application context, baseline comparisons, RES configurations, BEMS types, building typologies, and exogenous signals such as prices, weather, occupancy, renewable generation, and battery state of charge.	The survey contributes to RQ6 mainly by making implementation attributes and external-signal assumptions visible through structured summary tables, but it remains weaker than the strongest RQ6 papers on open-source practices, benchmark infrastructure, and explicit reproducibility critique.

Table S6.102: One-line RQ6 answer for Michailidis et al. (2025), after recoding.

Survey	RQ6 answer
Michailidis et al. (2025)	Michailidis et al. (2025) answer RQ6 mainly by showing that RL-based renewable-energy building control is shaped by application context, baseline choice, BEMS and RES configuration, and strong dependence on external signals such as weather, pricing, occupancy, renewable generation, and battery state, while giving only limited attention to shared benchmarks, open code, or artifact reproducibility.

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